B.TECH CAPSTONE AI & DS CSIT 2026–27

MACHINE LEARNING PROJECT

Credit Card Fraud Detection Using Logistic Regression and Isolation-Forest Anomaly Detection

PROJECT INVESTIGATORS

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PROJECT GUIDE

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Introduction

CHALLENGE & EVOLUTION

- The increasing use of online banking and digital payment systems has increased the importance of credit card fraud detection.
- Traditional fraud detection approaches can struggle with new and evolving fraud patterns.

MACHINE LEARNING SOLUTION

- Machine learning can identify suspicious transaction patterns using data-driven techniques.
- The proposed system uses two complementary approaches: Logistic Regression and Isolation Forest.

VISUAL FLOW: DIGITAL TRANSACTIONS → TRANSACTION DATA → MACHINE LEARNING → FRAUD DETECTION

Digital Transactions

Transaction Data

Machine Learning

Fraud Detection

Problem Statement & Objectives

Detect fraudulent credit card transactions accurately while identifying both known and previously unseen fraud patterns.

CORE PROJECT OBJECTIVES

- Detect fraudulent transactions efficiently
- Classify known fraud patterns
- Identify unusual and emerging transaction behavior
- Reduce false positives and false negatives
- Improve the reliability and speed of fraud detection
- Support safer digital payment systems

DETECTION PARADIGM COMPARISON

TRADITIONAL DETECTION

Limited Adaptability

Rigid static rules, high operational overhead, fails against novel attack vectors.

MACHINE LEARNING

Data-Driven & Adaptive Detection

Dynamically learns feature relationships and flags statistical anomalies.

Proposed Machine Learning Approach

SUPERVISED CLASSIFICATION

LOGISTIC REGRESSION

- Supervised learning algorithm
- Uses labeled transaction data
- Classifies transactions as legitimate or fraudulent
- Learns relationships between transaction features and fraud labels

UNSUPERVISED ANOMALY DETECTION

ISOLATION FOREST

- Unsupervised anomaly detection algorithm
- Identifies unusual transaction behavior
- Detects data points that differ significantly from normal behavior
- Can identify previously unseen or emerging fraudulent activities

System Workflow

01

Data Ingestion

1. Credit Card Dataset
2. Data Preprocessing
3. Feature Scaling

02

Model Pipeline

1. Handling Class Imbalance
2. Model Training
3. Logistic Regression + Isolation Forest

03

Inference & Verification

1. Testing
2. Performance Evaluation
3. Fraud / Legitimate Transaction Detection

Data Preprocessing

- Data cleaning and preparation
- Feature scaling
- Handling class imbalance
- Preparing labeled data for Logistic Regression
- Preparing transaction data for anomaly detection
- Splitting data for training and testing

PIPELINE: RAW TRANSACTION DATA → CLEANED DATA → SCALED FEATURES → BALANCED/PREPARED DATASET → ML MODELS

1
Raw Transaction Data

2
Cleaned Data

3
Scaled Features

4

Model Training & Fraud Detection

Combined Insights → Comprehensive Fraud Detection

SUPERVISED INFERENCE TRACK

LOGISTIC REGRESSION

LABELED TRANSACTIONS → MODEL TRAINING → FRAUD PROBABILITY/CLASS → FRAUD OR LEGITIMATE

Learns the hyper-plane separating known legitimate transactions from fraudulent ones by assigning a calibrated probability score.

UNSUPERVISED ISOLATION TRACK

ISOLATION FOREST

TRANSACTION FEATURES → ISOLATION PROCESS → ANOMALY SCORE → NORMAL OR SUSPICIOUS

Constructs an ensemble of isolation trees to identify rare instances requiring fewer partition splits than standard transaction points.

Supervised classification handles known fraud patterns while anomaly detection helps identify unusual and emerging patterns.

Performance Evaluation

KEY EVALUATION CRITERIA ESTABLISHED FOR HEAVILY IMBALANCED CREDIT CARD TRANSACTION DATASETS.

ACCURACY

Accuracy

Overall classification correctness

PRECISION

Precision

Correctness of fraud predictions

RECALL

Recall

Ability to identify fraudulent transactions

F1-SCORE

F1-Score

Balance between precision and recall

ROC-AUC

ROC-AUC

Overall classification performance across thresholds

NO NUMERICAL MODEL RESULTS ARE PROVIDED IN THE PROJECT SOURCE.

Technology Stack & Requirements

DEVELOPMENT

Programming & Development

- Python 3.10+
- Jupyter Notebook / Visual Studio Code

FRAMEWORKS

Machine Learning & Data Processing

- Scikit-learn
- Pandas
- NumPy

ANALYTICS

Visualization & Model Saving

- Matplotlib
- Seaborn
- Joblib

INFRASTRUCTURE

Dataset & Hardware

- Credit Card Fraud Detection Dataset (CSV)
- Intel Core i3 or equivalent; 4 GB RAM minimum; 8 GB recommended; 500 MB free storage; Internet connection

Conclusion

PROJECT SUMMARY & IMPACT

- The project combines supervised and unsupervised machine learning techniques for credit card fraud detection.
- Logistic Regression detects known fraud patterns using labeled transaction data.
- Isolation Forest identifies unusual and potentially emerging fraudulent behavior.
- The approach aims to improve fraud detection reliability and speed.
- The system can help reduce financial losses and improve digital payment security.
- The solution has potential applications for banks, financial organizations, and online payment platforms.

VIVA EVALUATION

Thank You

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